

Demonstration Coverage and Extrapolation

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The Claim

A model that learns from demonstrations fails on hard problems not because it lacks the capability, but because the demonstrations did not cover that difficulty. Adding one demonstration at the harder level restores performance. The model weights do not change. The question does not change. Only the examples change.

The Task

We use an ordering task: colored bars on a 10×10 grid, sorted by height. Complexity is the number of bars (2 to 8). Each task set has three demonstration pairs and one query.

Covered

At least one demo matches the query complexity.

Uncovered

All demos have at most 3 bars. The query grid is identical in both conditions. Only the demonstrations differ.

The Model

We trained a decoder-only transformer (6 layers, 256 dimensions, 8 attention heads, ~4.2 million parameters) on 10,500 covered tasks across all complexities. With the right demonstrations present, the model learned to solve every difficulty level. At inference, we do not retrain or update the model. We change only the demonstrations it reads as input context.



Results

Complexity	Covered EM	Uncovered EM	Gap
2	100%	96%	4 pp
3	100%	0%	100 pp
5	100%	0%	100 pp
8	100%	0%	100 pp

EM is exact match: the output grid matches cell for cell. The model capacity and weights are fixed. Success depends on whether the input context contains a demonstration at the right difficulty.

The Meta-Learning Connection

In-context learning is implicit meta-learning. Von Oswald et al. (arXiv:2212.07677, ICML 2023) showed that a transformer forward pass implements gradient-descent-style weight updates on its internal representations. The demonstrations are not passive context. They are a training signal processed in a single pass.

Min et al. (arXiv:2202.12837, EMNLP 2022) found that the distribution of demonstrations matters more than label correctness. This predicts our finding: what breaks in-context learning is not wrong examples but missing difficulty levels.

The same problem appears at frontier scale. GPT-6 Astra (OpenAI, Sep 2026) and Claude Opus 5 (Anthropic, 2026) both rely on in-context learning. Our 4.2M toy model makes the effect visible. Frontier models make it consequential.

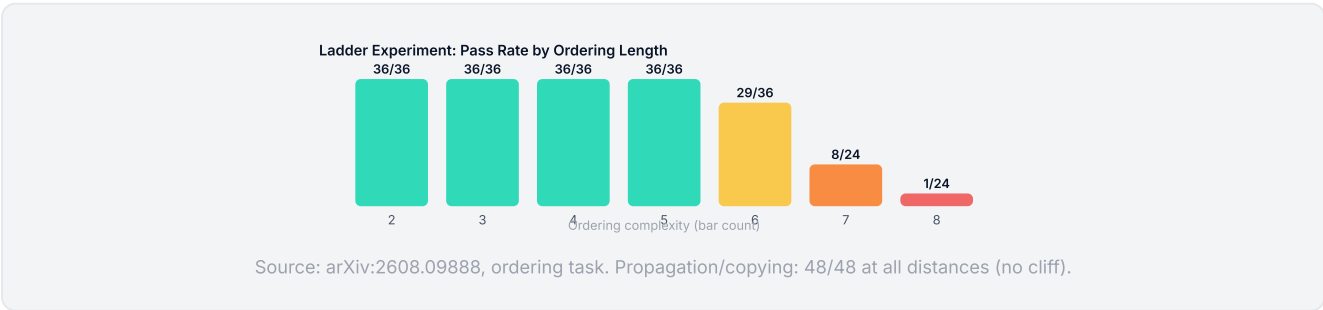
Connection to BDH-CQ

BDH-CQ is a 150M-parameter model built on Pathway's Brain-inspired Dragon Hatchling architecture (arXiv:2608.09888). It stores demonstrations in a recurrent synaptic state instead of a growing key-value cache. It scored 29.5% pass@2 on ARC-AGI-1 at \$0.00070 per task.

Task	Uncovered	Covered	Source
Ordering, length 8	0/24	12/24 (pass@1)	arXiv:2608.09888, Table 3
Nesting, depth 5	15/24	16/24 (pass@1)	arXiv:2608.09888, Table 3

The ladder experiment shows a gradual falloff for ordering: accuracy stays high through complexity 5, then falls to 29/36 at 6, 8/24 at 7, 1/24 at 8. Propagation and copying tasks show no falloff (48/48 at all distances).

Some operations depend on demonstration coverage more than others.



System Comparison

System	Adaptation method	Weight updates	Memory	Cost per task
BDH-CQ	Recurrent state absorbs demos	No	Fixed-size matrix	\$0.00070
HRM	Gradient optimization on demo pairs	Yes	Growing buffers	\$1.48
TRM	Learned identity embedding per puzzle	Yes	Growing buffers	\$1.76

Costs: independent audit by co-authors at Bielik and NYU (arXiv:2608.09888). HRM: arXiv:2506.21734. TRM: 7M params, 45% ARC-AGI-1.

Coverage sensitivity in ICL is an active research area. Brown et al. (GPT-3, 2020) showed that few-shot performance scales with example count but did not isolate complexity matching as the failure variable.

Strengths and Limitations

The claim is falsifiable and reproducible in under 60 seconds. The artifact runs a real model, not an animation. Every number traces to a paper.

Our model trains on a single synthetic task family. The coverage cliff is specific to ordering. Other operations (like propagation) do not show this cliff, even in BDH-CQ results.

The Hebbian memory visualization is a simplified illustration of the BDH update rule (Pathway BDH Explainer, Chapter 2). It omits low-rank compression, the positional operator, and gating.

All BDH-CQ numbers are developer-reported. No public model weights exist. We did not run BDH-CQ. The system dimensions and update rules remain proprietary. No independent team reproduced the results. BDH-CQ is a research prototype, not a production deployment.

Demonstration adaptation is session-scoped. The model does not learn across tasks or retain knowledge from one session to the next. Consolidating fast synaptic state into durable weights remains an open problem (Dragon Hatchling, arXiv:2509.26507, Conclusion).

Key References

BDH-CQ report, arXiv:2608.09888, August 2026

Dragon Hatchling, arXiv:2509.26507, September 2025

Coconut (latent reasoning baseline), arXiv:2412.06769, December 2024

Von Oswald et al., "Transformers Learn In-Context by Gradient Descent," arXiv:2212.07677, ICML 2023

Min et al., "Rethinking the Role of Demonstrations," arXiv:2202.12837, EMNLP 2022